Arya Akhavan
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Experience

- **University of Oxford**
  Postdoctoral Researcher; working with Patrick Rebeschini — *Sep. 2024 – present*
- **University of Alberta**
  Visiting Researcher; working with Csaba Szepesvári — *Mar. 2026 – May 2026*
- **Vector Institute and University of Toronto**
  Visiting Researcher; working with Murat A. Erdogdu — *May 2026*
- **École Polytechnique de Paris**
  Postdoctoral Researcher; working with Karim Lounici and El Mahdi El Mhamdi — *Mar. 2023 – Jul. 2024*

Education

- **ENSAE, Institut Polytechnique de Paris and IIT-Genova**
  PhD in Theoretical Machine Learning — *Sep. 2019 – Feb. 2023*
  - **Advisors**: Alexandre B. Tsybakov and Massimiliano Pontil
- **Paris Dauphine University and École Normale Supérieure Paris-Saclay**
  Master M2 MASH Learning and Applications; Mention: Très Bien — *Sep. 2017 – Dec. 2018*
- **Sharif University of Technology**
  First year of Master in Mathematics; Mention: Très Bien — *Sep. 2016 – Jul. 2017*
- **University of Tehran**
  B.Sc. in Mathematics — *Sep. 2011 – Jul. 2016*

Research interests

- **Memory-augmented reasoning; AI safety; reinforcement and imitation learning; online learning and optimization.**

Work in Progress

- **A. Akhavan**, D. Janz, P. Rebeschini, and C. Szepesvári (2026a). "External memory for reliable long-horizon reasoning and state tracking". Manuscript in preparation
- A. Masoumian, **A. Akhavan**, and C. Szepesvári (2026). "Learning when to ask: Adaptive expert queries for imitation learning". Manuscript in preparation
- **A. Akhavan**, M. A. Erdogdu, P. Rebeschini, and C. Szepesvári (2026c). "Memory-augmented AI safety: Learning when and how to intervene in language model reasoning". Ongoing research project on using memory-based interventions to improve the safety and reliability of language-model reasoning
- **A. Akhavan**, D. Gomez, P. Rebeschini, and C. Szepesvári (2026b). "From past solutions to future plans: Reasoning across tasks with growing memory". Ongoing research project on reusing and expanding memories of prior solutions to plan across tasks

Publications and Preprints

- D. Janz, **A. Akhavan**, and A. B. Tsybakov (2026). "Maximum effective dimension and information gain". Under revision. arXiv: 2608.24450
- A. Carpentier, C. Rouyer, A. B. Tsybakov, and **A. Akhavan** (2026). "Sharp optimal algorithm for derivative-free stochastic convex optimization in one dimension". Under revision. arXiv: 2607.12938
- **A. Akhavan**, S. Louati, and A. B. Tsybakov (2026). "Gradient-free stochastic optimization of derivatives under strong convexity". Under revision. arXiv: 2607.07249
- D. Janz, **A. Akhavan**, and C. Szepesvári (2026). "Sharp analysis of linear ensemble sampling". In: *Proceedings of the Thirty-Ninth Conference on Learning Theory (COLT)*. vol. 336. Proceedings of Machine Learning Research, pp. 3716–3750. arXiv: 2602.08026
- L. Lévy, J.-L. Valeau, **A. Akhavan**, and P. Rebeschini (2026). "Self-concordant perturbations for linear bandits". In: *Proceedings of the Thirty-Ninth Conference on Learning Theory (COLT)*. vol. 336. Proceedings of Machine Learning Research, pp. 4635–4673. arXiv: 2510.24187
- D. Janz, E.-M. El-Mhamdi, and **A. Akhavan** (2025). "High-probability zeroth-order online convex optimisation beyond Euclidean geometry". Preprint. arXiv: 2509.21484
- **A. Akhavan**, A. Shidani, A. Ayoub, and D. Janz (2025). "Bernstein-type dimension-free concentration for self-normalised martingales". Under revision. arXiv: 2507.20982
- X. Liu, D. Baudry, J. Zimmert, P. Rebeschini, and **A. Akhavan** (2025). "Non-stationary bandit convex optimization: A comprehensive study". In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 2506.02980
- **A. Akhavan** and A. B. Tsybakov (2025). "Gradient-free stochastic optimization for additive models". In: *Automation and Remote Control* 86.9, pp. 817–830. arXiv: 2503.02131
- **A. Akhavan**, E. Chzhen, M. Pontil, and A. B. Tsybakov (2024b). "Gradient-free optimization of highly smooth functions: Improved analysis and a new algorithm". In: *Journal of Machine Learning Research* 25.370, pp. 1–50
- **A. Akhavan**, K. Lounici, M. Pontil, and A. B. Tsybakov (2024a). "A conversion theorem and minimax optimality for continuum contextual bandits". Under revision. arXiv: 2406.05714
- **A. Akhavan**, D. Gogolashvili, and A. B. Tsybakov (2024). "Estimating the minimizer and the minimum value of a regression function under passive design". In: *Journal of Machine Learning Research* 25.11, pp. 1–37
- R. Grazzi, **A. Akhavan**, J. I. T. Falk, L. Cella, and M. Pontil (2022). "Group meritocratic fairness in linear contextual bandits". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 35
- **A. Akhavan**, E. Chzhen, M. Pontil, and A. B. Tsybakov (2022). "A gradient estimator via L1-randomization for online zero-order optimization with two-point feedback". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 35
- **A. Akhavan**, M. Pontil, and A. B. Tsybakov (2021). "Distributed zero-order optimization under adversarial noise". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 34
- **A. Akhavan**, M. Pontil, and A. B. Tsybakov (2020). "Exploiting higher order smoothness in derivative-free optimization and continuous bandits". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 33

Service

- **Area Chair**
  International Conference on Machine Learning (ICML) *2027*
- **Organizer of Seminars**
  Weekly seminars at the University of Oxford, Oxford, United Kingdom *November 2024 - present*
- **Co-organizer of a Conference**
  Re-thinking high dimensional statistics, Oberwolfach, Germany *May 2022*
- **Instructor of Problem Solving Sessions for the Team of University of Tehran**
  To prepare for 40th Iranian Mathematical Society Competition *May 2016*
- **Co-head of the Mathematical Olympiad Team of University of Tehran**
  In 40th Iranian Mathematical Society Competition, Iran University of Science and Technology. *Aug. 2016*

Teaching Experience

- **Reinforcement Learning**, 6 hours, developed and instructed by myself at University of Oxford. *Feb. 2025*
- **Reinforcement Learning**, 21 hours, developed and instructed by myself at Paris Dauphine University. *Feb. 2023, Oct. 2023*
- **Tutorial on Bandit Algorithms**, 5 hours, developed and instructed by myself at École Polytechnique de Paris. *Feb. 2023, Oct. 2023*
- **High Dimensional Statistics**, 12 hours, instructed by Evgenii Chzhen at ENSAE, IP, Paris. *Oct. 2023*
- **Online Learning**, 12 hours, instructed by Alexandre B. Tsybakov at ENSAE, IP, Paris. *Feb. 2023*
- **Number Theory**, 32 hours, instructed by Amir Ghadermarzi at University of Tehran. *Aug. 2017*
- **Topology**, 32 hours, instructed by Hadi Zare at University of Tehran. *Sep. 2016*
- **Linear Algebra**, 32 hours, instructed by Hadi Zare at University of Tehran. *Sep. 2016*

Honors and Awards

- **Top 10% of Reviewers**, Neural Information Processing Systems (NeurIPS) *2022 - 2024*
- **Top 10% of Reviewers**, International Conference on Machine Learning (ICML) *2022 - 2024*
- **Ranked 1st**, evaluation of the master's thesis by the committee of Master M2 MASH in Paris Dauphine University *2018*
- **Ranked 3rd**, overall coursework in Master M2 MASH *2018*
- **Granted by Fondation Sciences Mathématiques de Paris** *2017*
- **Silver Medal**, in 39th Iranian Mathematical Society Competition. *2015*

Skills and Languages

- **Programming & Scientific Computing**: Python, NumPy.
- **Deep Learning**: PyTorch, neural network implementation, transformer architectures.
- **LLM Fine-Tuning**: Supervised fine-tuning (SFT), parameter-efficient fine-tuning (LoRA/PEFT).
- **Languages**: Persian (native), Armenian (native), English (fluent), French (working proficiency)